ICSE 2027 EXAMINATION

CHAPTER-WISE QUESTION BANK

CHEMISTRY (SCIENCE PAPER – 2)

Class X · Chapters 1, 2 and 3 · Format of the 2027 Specimen Paper

About this bank: Every question is taken from the Chemistry papers in your files (board papers 2023–2026 and specimen papers 2023–2027). Nothing has been added. The grey tag after each question shows its source (T = board paper, S = specimen paper, with the year). Marks are shown in brackets, and questions are grouped by type: Section A (Question 1 multiple choice, Question 2 short parts) and Section B (structured questions).

CHAPTER 1: PERIODIC PROPERTIES AND VARIATIONS OF PROPERTIES

SECTION A

(Attempt all questions from this Section.)

Question 1 [20]

Choose the correct answers to the questions from the given options. (Do not copy the question, write the correct answers only.)

(i) An element in period 3, whose electron affinity is zero: [T23 Q1(i)]

(a) Neon
(b) Sulphur
(c) Sodium
(d) Argon

(ii) An element with the largest atomic radius among the following is: [T23 Q1(ii)]

(a) Carbon
(b) Nitrogen
(c) Lithium
(d) Beryllium

(iii) In the 2nd period Neon has maximum Ionization Potential because: [T24 Q1(ii)]

(a) It has unstable electronic configuration.
(b) It easily accepts electrons.
(c) It easily loses electrons.
(d) The outer most shell is completely filled.

(iv) Electron Affinity is maximum in: [T24 Q1(viii)]

(a) Mg
(b) Ar
(c) Li
(d) Br

(v) Assertion (A): The tendency of losing electrons increases down the Group. [T25 Q1(vi)]

Reason (R): The most reactive metal is placed at the top of Group 1.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, and (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

(vi) Given below are four ions: Cl^-, Li^+, Al^{3+}, K^+. Identify the pair of ions which have the same electronic configuration. [Atomic number: Cl = 17, Li = 3, Al = 13, K = 19] [T25 Q1(xi)]

(a) Cl^- & Li^+
(b) Al^{3+} & K^+
(c) Cl^- & K^+
(d) Li^+ & K^+

(vii) X, Y & Z are three metallic atoms in successive order belonging to the same group such that atomic radii of 'X' is the smallest. Which of the three atoms is the best reducing agent? [T25 Q1(xiv)]

(a) X
(b) Y
(c) Z
(d) All three have the same reducing power.

(viii) An atom of X forms an ion according to the equation $X \rightarrow X^{+2}$. The atomic number of the atom X is: [T26 Q1(viii)]

(a) 16
(b) 10
(c) 12
(d) 14

(ix) Electron affinity is maximum in: [S23 Q1(ii)]

(a) Alkaline earth metals
(b) Halogens
(c) Inert gases
(d) Alkali metals

(x) An alkali metal found in period 3 and group 1 is: [S23 Q1(xv)]

(a) Magnesium
(b) Lithium
(c) Sodium
(d) Potassium

(xi) $A \rightarrow A^{+3}$; $B \rightarrow B^{-1}$. Number of electrons present in the outermost shell of atoms A and B respectively are: [S24 Q1(viii)]

(a) 5, 1
(b) 3, 1
(c) 3, 7
(d) 5, 7

(xii) An element with atomic no. ___ will form an acidic oxide. [S24 Q1(x)]

(a) 3
(b) 17
(c) 11
(d) 13

(xiii) An element belongs to third period and sixteenth group. It will have ___ electrons in its valence shell. [S24 Q1(xv)]

(a) 2
(b) 5
(c) 6
(d) 3

(xiv) Element 'P' has electronic configuration 2, 8, 8, 1. The number of chlorine atoms present in the chloride of 'P' is: [S25 Q1(vii)]

(a) 2
(b) 1
(c) 3
(d) 4

(xv) $_1H^2$ is an isotope of hydrogen. In the modern Periodic Table it will: [S25 Q1(viii)]

(a) be placed before hydrogen
(b) be placed after hydrogen
(c) be placed at the same position as hydrogen
(d) not have any position in the Periodic Table

(xvi) Which of the following electronic configuration represents the most electropositive element? [S25 Q1(x)]

(a) 2, 1
(b) 2, 8, 1
(c) 2, 2
(d) 2, 8, 2

(xvii) Assertion (A): Alkali metals do not form dipositive ions. [S25 Q1(xi)]

Reason (R): After loss of one electron alkali metals achieve stable electronic configuration of noble gases.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

(xviii) Which element amongst the following has the largest atomic radius? [S26 Q1(x)]

(a) Al
(b) S
(c) Mg
(d) Na

(xix) X and Y are diatomic elements. X is less reactive than Y. What are elements X and Y? [S27 Q1(i)]

(a) X - chlorine, Y - iodine
(b) X - fluorine, Y - nitrogen
(c) X - iodine, Y - bromine
(d) X - oxygen, Y - nitrogen

(xx) Statement I: Halogens are very reactive. [S27 Q1(vii)]

Statement II: Halogens have high electron affinity.

(a) Both the statements are true.
(b) Both the statements are false.
(c) Statement I is true, and statement II is false.
(d) Statement I is false, and statement II is true.

Question 2 [5]

(i) Identify the following: **[3]**

(a) The energy released when an atom in the gaseous state accepts an electron to form an anion. [T23 Q2(iv)(a)]

(b) The tendency of an atom to attract the shared pair of electrons towards itself when combined in a compound. [T24 Q2(iv)(d); T25 Q2(ii)(d)]

(c) The energy required to remove an electron from a neutral gaseous atom. [S23 Q2(iv)(c); S24 Q2(iv)(d); S25 Q2(iv)(d); S26 Q2(ii)(d)]

(ii) Complete the following by choosing the correct answers from the brackets: **[2]**

(a) If an element has one electron in the outermost shell, then it is likely to have the ___ [smallest / largest] atomic size amongst all the elements in the same period. [S24 Q2(iii)(a); S25 Q2(iii)(a)]

(b) The electron affinity of element X is greater than that of element Y. The oxidising power of X is likely to be ___ [more / less] than that of element Y. [S26 Q2(iii)(d)]

SECTION B

(Attempt any four questions from this Section in the paper; all questions below are for practice.)

Question 3

(i) Arrange the following as per the instruction given in the brackets: **[3]** [T23 Q3(iii)]

(a) Al, K, Mg, Ca (decreasing order of its reactivity)

(b) N, Be, O, C (increasing order of non-metallic character)

(c) P, Si, F, Be (decreasing order of valence electrons)

(ii) Give reason for the following: **[2]** [T23 Q7(ii)]

(a) Ionisation potential decreases down a group.

(b) Ionic compounds do not conduct electricity in solid state.

Part (b) belongs to Chapter 2.

(iii) Arrange the following as per the instructions given in the brackets: **[3]** [T24 Q6(iii)]

(a) Carbon, Fluorine, Beryllium (decreasing order of atomic size)

(b) Sulphuric acid, Phosphoric acid, Acetic acid (increasing order of number of replaceable H atoms per molecule)

(c) Potassium, Lithium, Sodium (increasing order of ionization potential)

Part (b) belongs to Chapter 3.

(iv) Identify the following: **[3]** [T24 Q6(iv)]

(a) An element in period 1 which can be placed in both group 1 and group 17 of the Periodic Table.

(b) The element having electronic configuration 2, 8, 6.

(c) The most electronegative element of period 3.

Question 4

(i) The atomic number of two atoms 'X' and 'Y' are 14 and 8 respectively. State: **[2]** [T25 Q3(i)]

(a) the period to which 'X' belongs.

(b) the formula of the compound formed between 'X' and 'Y'.

(Do not identify X and Y)

(ii) Given below are two sets of elements from two different periods. Name the element with the highest ionisation potential in each of the following sets. **[2]** [T25 Q5(i)]

(a) Al, Cl, Mg

(b) Ne, O, F

(iii) Elements P, Q, and R are in the same period of the modern periodic table. **[3]** [T26 Q4(iii)]

- P readily loses its one valence electron to form a stable ion.
- Q shares its electrons in bonding but does not form ions easily.

• R has high electronegativity.

Answer the following questions based on the above information:

(a) Which element would be most difficult to reduce among P, Q and R?

(b) Which element is expected to have the smallest atomic radius among P, Q and R?

(c) Arrange P, Q and R in order of decreasing ionization potential.

(iv) Name the following: **[2]** [T26 Q8(ii)]

(a) The most electronegative element of Period 2.

(b) The largest atom of Period 3.

Question 5

(i) Given below are a few elements. **[3]** [T26 Q8(iii)]

Li, K, Ca, F

Identify the element which:

(a) has the least atomic radius.

(b) has two valence electrons.

(c) is the most electropositive.

(ii) Arrange the following as per the instruction given in the brackets: **[3]** [S23 Q3(iii)]

(a) Na, K, Cl, Si, S (increasing order of electronegativity)

(b) Be, Li, F, C, B, N, O (increasing order of metallic character)

(c) Br, F, I, Cl (increasing order of atomic size)

(iii) An element X has atomic number 17. Answer the following questions. **[3]** [S23 Q8(iv)]

(a) State the period & group to which it belongs.

(b) Is it a Metal or Non Metal?

(c) Write the formula between X and Hydrogen.

(iv) The electron affinity of an element X is greater than that of element Y. **[3]** [S24 Q3(iii); S25 Q3(iii); S26 Q3(iii)]

(a) How is the oxidising power of X likely to compare with that of Y?

(b) How is the electronegativity of X likely to compare with that of Y?

(c) State whether X is likely to be placed to the left or to the right of Y in the periodic table?

Question 6

(i) X [2, 8, 7] and Y [2, 8, 2] are two elements. Using this information complete the following: **[3]** [S24 Q6(iv); S25 Q6(iv)]

(a) ___ is the metallic element.

(b) Metal atoms tend to have a maximum of ___ electrons in the outermost shell.

(c) ___ is the reducing agent.

(ii) Element 'X' forms an oxide with the formula X_2O_3 which is a solid with high melting point. 'X' would most likely be placed in the group of the Periodic Table as: **[2]** [S25 Q6(i)]

1. (a) Na (b) Mg (c) Al (d) Si

2. Justify your answer in the above question (1).

(iii) L, M and N are three elements with atomic numbers 13, 7 and 10 respectively. Answer the following questions using only the alphabets given. Do not identify the elements. **[3]** [S26 Q8(iv)]

Which element:

(a) can combine with hydrogen to form a gas which produces dense white fumes with concentrated HCl?

(b) has zero electron affinity?

(c) can form an ionic compound with oxygen?

(iv) An element 'P' has 3 valence electrons in its L-shell and an element 'Q' has 7 valence electrons in its M-shell. Using this information, answer the following questions: **[3]** [S27 Q5(iv)]

(a) Identify the exact position of element 'Q' in terms of group and period in the Modern Periodic Table.

(b) The atomic radius of P is ___ (more / less) than that of Q.

Question 7

(i) In the reactivity series of metals, Z is a metal above hydrogen and its oxide has the formula ZO. It was found that ZO reacted with water and formed a hydroxide as the product. With reference to the above context, answer the following questions: **[3]** [S27 Q8(i)]

(a) State the number of valence electrons present in element Z.

(b) Name the group which Z belongs to.

(c) What will be the formula of the hydroxide of Z?

CHAPTER 2: CHEMICAL BONDING

SECTION A

(Attempt all questions from this Section.)

Question 1 [6]

Choose the correct answers to the questions from the given options. (Do not copy the question, write the correct answers only.)

(i) In the molecule of water, oxygen atom has: [T24 Q1(xiv)]

(a) One shared pair of electrons.
(b) Three shared pairs of electrons.
(c) Two lone pairs of electrons.
(d) One lone pair of electrons.

(ii) Which of the following element pairs will form an ionic bond? [T26 Q1(ii)]

P – Elements of Group 1 & Group 2

Q – Elements of Group 14 & Group 16

R – Elements of Group 2 & Group 17

S – Elements of Group 15 & Group 18

(a) P
(b) Q
(c) R
(d) S

(iii) The compound that has the highest melting point amongst the following is: [T26 Q1(x)]

(a) Methane
(b) Sodium chloride
(c) Ammonia
(d) Ethanol

(iv) A polar covalent compound is: [S23 Q1(iv)]

(a) Methane
(b) Ammonia
(c) Nitrogen
(d) Chlorine

(v) Identify the ion that contains one lone pair of electrons. [S26 Q1(iv); S27 Q1(xii)]

(a) OH^{-1}
(b) H_3O^+
(c) NH_4^+
(d) H^+

(vi) Assertion (A): Ionic compounds have high melting points. [S27 Q1(viii)]

Reason (R): A strong electrostatic force of attraction exists between the oppositely charged ions.

(a) (A) is true but (R) is false.

(b) (A) is false but (R) is true.

(c) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(d) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

Question 2 [15]

(i) Identify the following: **[3]**

(a) A bond formed between two atoms by sharing of a pair of electrons, with both electrons being provided by the same atom. [S23 Q2(iv)(e); S24 Q2(iv)(a); S25 Q2(iv)(a); S26 Q2(ii)(a)]

(b) A pair of electrons present in an atom which is not shared with any other atom during bond formation. [T26 Q2(ii)(d)]

(c) The type of covalent bond in which electrons are shared unequally between the combining atoms. [T25 Q2(ii)(e)]

(ii) Complete the following by choosing the correct answers from the brackets: **[2]**

(a) Non-polar covalent compounds are ___ [good / bad] conductors of heat and electricity. [T23 Q2(iii)(b)]

(b) ___ [HCl / CCl_4] is a polar covalent compound. [T26 Q2(iv)(e)]

(iii) Match the Column A with Column B: **[5]** [T24 Q2(ii)]

Column A	**Column B**
(a) Water	1. Lithium
(b) Alkali metal	2. Iodine
(c) Halogen	3. Covalent compound
(d) Calcium oxide	4. Acetic acid
(e) Weak acid	5. Ionic compound
	6. Sulphuric acid

Mixed: also covers Chapters 1 and 3.

(iv) Match the Column A with Column B: **[5]** [T23 Q2(i)]

Column A	**Column B**
(a) Sodium Chloride	1. has two shared pair of electrons
(b) Methane	2. has high melting and boiling points
(c) Hydrogen chloride gas	3. a greenhouse gas
(d) Oxidation reaction	4. has low melting and boiling points
(e) Water	5. $Zn - 2e^- \rightarrow Zn^{2+}$
	6. $S + 2e^- \rightarrow S^{2-}$

Mixed: part (d) belongs to Chapter 6.

SECTION B

(Attempt any four questions from this Section in the paper; all questions below are for practice.)

Question 3

(i) Draw the electron dot structure of: **[2]** [T24 Q4(ii)]

(a) Methane molecule

(b) Nitrogen molecule

[Atomic number: N = 7, C = 6, H = 1]

(ii) Give reasons for the following: **[2]** [T24 Q6(ii)]

(a) Inert gases do not form ions.

(b) Covalent compounds have a low melting and boiling point.

(iii) Fill in the blanks by choosing the correct answer from the bracket: **[2]** [T24 Q7(ii)]

(a) Carbon tetrachloride is a ___ [polar / non-polar] covalent molecule.

(b) During electrolysis of acidulated water, the gas liberated at the anode is ___ [oxygen / hydrogen].

Part (b) belongs to Chapter 6.

(iv) Draw the electron dot structure for the following: **[2]** [T23 Q5(ii)]

(a) Ammonium ion

(b) A molecule of nitrogen

[At. No.: N = 7, H = 1]

Question 4

(i) Draw the dot and cross structure of the following: **[3]** [T25 Q7(iii)]

(a) An ionic compound formed when Mg reacts with the dil. HCl.

(b) A covalent compound formed when H_2 reacts with Cl_2.

(c) The positive ion produced when ammonia gas is dissolved in water.

[Atomic number: Mg = 12, Cl = 17, H = 1, N = 7]

(ii) (a) State Avogadro's Law. **[2]** [T25 Q8(i)]

(b) Define Co-ordinate bond.

Part (a) belongs to Chapter 5.

(iii) Fill in the blanks selecting the appropriate word from the given choice: **[3]** [S23 Q3(iv)]

(a) In a covalent compound the bond is formed due to ___ of electrons. (sharing / transfer)

(b) A molecule which has a single lone pair of electrons ___. (NH_3 / H_2O)

(c) Electrovalent compounds do not conduct electricity in their ___ state. (molten / solid)

(iv) Draw the electron dot structure for the following. **[2]** [S23 Q8(i)]

(a) H_3O^+

(b) CH_4

Question 5

(i) (a) State whether the following statements are TRUE or FALSE. Justify your answer. **[3]** [S24 Q3(iv)]

1. In an electrovalent compound, the cation attains the electronic configuration of the noble gas that comes after it in the periodic table.

2. In the formation of a compound PQ_2, atom P gives one electron to each atom of Q. The compound PQ_2 is a good conductor of electricity.

(b) Calculate the number of moles in 22 grams of carbon dioxide.

Part (b) belongs to Chapter 5. Official marking scheme not available in your files.

(ii) Draw the electron dot diagram of Hydronium ion. **[2]** [S24 Q8(ii)]

(iii) Draw the electron dot diagram of ammonium ion. [Atomic No.: N = 7, H = 1] **[2]** [S25 Q8(ii); S26 Q8(ii)]

(iv) A compound commonly used in the laboratory has formula H_2P. H represents hydrogen, P represents a non-metal. Draw a dot and cross structure for the formation of the compound H_2P. **[2]** [S27 Q3(i)]

Question 6

(i) The diagram below illustrates the structures of two atoms P and Q. **[2]** [S27 Q7(ii)]

What is the formula and mass of one mole of the compound formed when P and Q react together?

[At. wt of P is 23 and Q is 16]

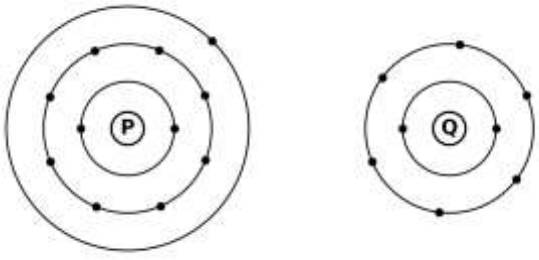


Diagram redrawn from the question data. Part of the answer is Chapter 5.

(ii) Draw the electron dot structure of: **[3]** [S27 Q8(ii)]

(a) ammonium ion (also label the co-ordinate bond)

(b) nitrogen molecule

[Atomic No.: N = 7, H = 1]

CHAPTER 3: STUDY OF ACIDS, BASES AND SALTS

SECTION A

(Attempt all questions from this Section.)

Question 1 [15]

Choose the correct answers to the questions from the given options. (Do not copy the question, write the correct answers only.)

(i) The electronic configuration of an element is 2, 8, 2. The hydroxide of this element can produce ___ hydroxyl ions per molecule. [T26 Q1(iii)]

(a) 3
(b) 2
(c) 1
(d) 0

(ii) Rita added dilute hydrochloric acid to four metals. She recorded her observations in the table given below. While noting her observations she made some errors. Which observation(s) made by Rita is/are correct? [T26 Q1(vii)]

Metals	Observations
1 copper	a gas was given off
2 iron	a gas was given off
3 magnesium	no gas was given off
4 zinc	a gas was given off

(a) 1 only
(b) 3 only
(c) Both 1 and 4
(d) Both 2 and 4

(iii) Salts can be prepared by the action of dilute acid on: 1. bases 2. carbonates 3. metals 4. sulphites. The method which cannot be used for the preparation of copper salts is: [T26 Q1(ix)]

(a) 1
(b) 2
(c) 3
(d) 4

(iv) Which pair of reactants can be best used to produce lead (II) sulphate? [T25 Q1(xii)]

(a) Sulphuric acid + Lead
(b) Sulphuric acid + Lead hydroxide
(c) Sodium sulphate + Lead nitrate
(d) Potassium sulphate + Lead oxide

(v) The acid which does not form acid salt by a basic radical: [T23 Q1(xiii)]

(a) H_2CO_3
(b) H_3PO_4
(c) H_2SO_4
(d) CH_3COOH

(vi) The indicator which changes to pink colour in an alkaline solution is: [T23 Q1(xv)]

(a) Blue Litmus
(b) Methyl Orange
(c) Red Litmus
(d) Phenolphthalein

(vii) An acid which has two replaceable hydrogen ions: [S23 Q1(v)]

(a) Acetic acid
(b) Hydrochloric acid
(c) Phosphoric acid
(d) Carbonic acid

(viii) An aqueous compound which turns colourless phenolphthalein to pink: [S23 Q1(x)]

(a) Ammonium hydroxide
(b) Nitric acid
(c) Anhydrous calcium chloride
(d) Sulphuric acid

(ix) The basicity of acetic acid is: [S24 Q1(vii)]

(a) 1
(b) 2
(c) 3
(d) 4

(x) Which of the following will dissociate in aqueous solution, to give a positive ion other than hydronium ion, and a negative ion other than hydroxyl ion? [S26 Q1(i)]

(a) KOH
(b) dil. HCl
(c) NaCl
(d) CH_3COOH

(xi) For which pH change is there the maximum increase in acidity? [S26 Q1(xi)]

Option	Initial pH	Final pH
(a)	1	3
(b)	2	6
(c)	3	1
(d)	6	2

(a) 1 to 3
(b) 2 to 6
(c) 3 to 1
(d) 6 to 2

(xii) When compound X reacts with dilute sulphuric acid, it releases a gas that turns acidified potassium dichromate solution from orange to green. Which of the following could be compound X? [S26 Q1(vi)]

(a) Lead nitrate
(b) Copper carbonate
(c) Sodium chloride
(d) Potassium sulphite

(xiii) A student tested the nature of four different solutions P, Q, R, and S using pH paper. He noted the following observations. Which of the above solutions is neutral in nature? [S27 Q1(ix)]

	P	**Q**	**R**	**S**
Colour of the pH paper after testing	Blue	Green	Red	Violet

(a) P
(b) Q
(c) R
(d) S

(xiv) The following statements are related to the properties of sulphuric acid: [S27 Q1(iv)]

1. Barium chloride reacts with sulphuric acid to give a white precipitate.

2. Sulphur dioxide gas is evolved when sulphuric acid reacts with carbon.

3. Sulphuric acid has a pH value of less than 7.0.

4. Copper (II) oxide reacts with sulphuric acid to form a blue solution.

Which two statements confirm the acidic nature of sulphuric acid?

(a) 1 and 2
(b) 1 and 3
(c) 3 and 4
(d) 2 and 4

(xv) When compound X reacts with dilute hydrochloric acid, it releases a gas that turns moist lead acetate paper black and decolourises acidified potassium permanganate solution. Which of the following is compound X? [S27 Q1(xiii)]

(a) Sodium nitrate
(b) Sodium sulphide
(c) Sodium carbonate
(d) Sodium sulphite

Question 2 [12]

(i) Identify the following: **[2]**

(a) A salt formed by the partial replacement of the hydroxyl group of a di-acidic or a tri-acidic base by an acid radical. [T23 Q2(iv)(d); T25 Q2(ii)(b)]

(b) A salt formed by the complete neutralization of an acid by a base. [S24 Q2(iv)(b); S25 Q2(iv)(b); S26 Q2(ii)(b)]

(ii) Complete the following by choosing the correct answers from the brackets: **[5]**

(a) Higher the pH value of a solution, the more ___ [acidic / alkaline] it is. [T23 Q2(iii)(c)]

(b) The salt that can be prepared by Direct Combination is ___ [$FeCl_3$ / $FeCl_2$]. [T24 Q2(iii)(a)]

(c) ___ [sulphuric acid / hydrochloric acid] does not form an acid salt. [S24 Q2(iii)(b); S25 Q2(iii)(b)]

(d) An ___ [alkaline / acidic] solution will turn methyl orange solution pink. [S24 Q2(iii)(e); S25 Q2(iii)(e)]

(e) A salt prepared by neutralization in which titration is involved is ___ [$CaCl_2$ / NH_4Cl]. [S27 Q2(iii)(e)]

(iii) Match the following Column A with Column B. **[5]** [S23 Q2(ii)]

Column A	**Column B**
(a) Acid Salt	1. Black in colour
(b) Copper Oxide	2. Reddish brown
(c) Zinc hydroxide	3. Hydrogen chloride
(d) Copper Metal	4. Sodium Hydrogen Carbonate
(e) Polar compound	5. Soluble in excess sodium hydroxide

Mixed: also covers Chapters 2 and 4.

SECTION B

(Attempt any four questions from this Section in the paper; all questions below are for practice.)

Question 3

(i) Differentiate between the following pairs based on the criteria given in the bracket: **[2]** [T26 Q4(i)]

(a) Acetic acid and Sulphuric acid (number of replaceable hydrogen ion per molecule)

(b) Electrolyte and Metallic conductor (the particles conducting electricity)

Part (b) belongs to Chapter 6.

(ii) Give reasons why: **[2]** [T26 Q6(i)]

(a) Hydrochloric acid cannot form an acid salt.

(b) Electronegativity increases across a period.

Part (b) belongs to Chapter 1.

(iii) State whether the following statements are True or False: **[2]** [T26 Q6(ii)]

(a) Organic compounds having different molecular formula, but same structural formula are called isomers.

(b) A salt is a compound formed by partial or complete replacement of the hydrogen ion of an acid by a metal or electro positive ion.

Part (a) belongs to Chapter 9.

(iv) Given below are some chemicals. **[3]** [T26 Q4(iv)]

Lead nitrate, Copper, Hydrochloric acid, Lead oxide,

Iron, Sodium carbonate, Copper carbonate, Chlorine

Using suitable chemicals from the box given above, write balanced chemical equations for the preparation of the following salts.

(Note: The chemicals may be used more than once, if needed.)

(a) Lead (II) carbonate

(b) Copper (II) chloride

(c) Iron (II) chloride

Question 4

(i) Match the salts underlined in Column A with the most suitable method of preparation given in Column B. **[3]** [T24 Q8(iii)]

Column A	**Column B**
(a) $ZnCl_2$ from Zn	1. Precipitation
(b) KNO_3 from KOH	2. Direct combination
(c) $CaCO_3$ from $CaCl_2$	3. Displacement reaction
	4. Neutralization

(ii) Mohan has three solutions P, Q and R having a pH of 13, 5 and 2 respectively. Which of the above solutions P, Q or R: **[3]** [T24 Q3(iii)]

(a) will react with Magnesium to liberate hydrogen gas?

(b) will liberate ammonia gas when it reacts with ammonium chloride?

(c) will contain molecules as well as ions?

(iii) Define the following terms: **[2]** [T24 Q4(i)]

(a) Molar volume

(b) Normal salt

Part (a) belongs to Chapter 5.

(iv) P, Q, R and S are the different methods of preparation of salts. **[3]** [T25 Q8(iv)]

P – Simple displacement

Q – Neutralisation by titration

R – Precipitation

S – Direct combination

Choose the most appropriate method to prepare the following salts:

(a) $PbCl_2$

(b) $FeCl_3$

(c) Na_2SO_4

Question 5

(i) Convert the following reactions into a balanced chemical equation: **[3]** [T23 Q8(iii)]

(a) Ammonia to nitric oxide using oxygen and platinum catalyst.

(b) Sodium hydroxide to sodium sulphate using sulphuric acid.

(c) Ferrous sulphide to hydrogen sulphide using hydrochloric acid.

Part (a) belongs to Chapter 8.

(ii) Choose the answer from the list which fits the description. **[3]** [S23 Q7(iv)]

[CaO, CO_2, NaOH, $Fe(OH)_3$, CO]

(a) A basic oxide.

(b) An oxide which is acidic.

(c) An Alkali.

(iii) Distinguish between the following as directed: **[2]** [S23 Q8(ii)]

(a) Sodium Carbonate and Sodium Sulphate by using dilute HCl

(b) Ammonium Sulphate and Sodium Sulphate by using Calcium hydroxide.

Part (b) also touches Chapter 4.

(iv) (a) Mahesh prepared a basic solution X that has a pH 7. How will the pH of the solution X change on addition of the following: **[3]** [S24 Q7(ii)]

1. Hydrochloric acid

2. a solution of a base

(b) The atomic number of an element is 15. To which group will this element belong to?

Question printed as in the paper (the solution is described as basic with pH 7). Official marking scheme not available in your files.

Question 6

(i) You are provided with the list of chemicals mentioned below in the box: **[3]** [S25 Q3(iv)]

Sodium hydroxide solution, copper carbonate, zinc,

hydrochloric acid, copper, dilute sulphuric acid, chlorine, iron

Using suitable chemicals from the list given, write balanced chemical equation for the preparation of the salts mentioned below:

(a) copper sulphate

(b) sodium zincate

(c) ferric chloride

(ii) You are provided with the list of chemicals mentioned below in the box: **[2]** [S26 Q6(ii)]

Sodium hydroxide solution, copper carbonate, zinc,

hydrochloric acid, copper, dilute sulphuric acid

Using suitable chemicals from the list given, write balanced chemical equation for the preparation of the salts mentioned below:

(a) copper sulphate

(b) sodium zincate

(iii) Rohit has solutions X, Y and Z that have pH 2, 7 and 13 respectively. Which solution out of X, Y or Z will: **[3]** [S25 Q7(ii); S26 Q7(iv); S27 Q4(ii)]

(a) liberate sulphur dioxide gas when heated with sodium sulphite?

(b) liberate ammonia gas when reacted with ammonium chloride?

(c) not have any effect on litmus paper?

ANSWER KEY

Official marking schemes were available in your files only for the specimen papers of 2025, 2026 and 2027. All other answers are worked out and should be checked against your textbook. For passage-based questions the passage is shown, followed by the answers only.

CHAPTER 1: PERIODIC PROPERTIES AND VARIATIONS OF PROPERTIES

Question 1

(i) (d) Argon

(ii) (c) Lithium

(iii) (d) The outermost shell is completely filled

(iv) (d) Br

(v) (c) (A) is true but (R) is false

(vi) (c) Cl^- & K^+

(vii) (c) Z

(viii) (c) 12

(ix) (b) Halogens

(x) (c) Sodium

(xi) (c) 3, 7

(xii) (b) 17

(xiii) (c) 6

(xiv) (b) 1

(xv) (c) be placed at the same position as hydrogen

(xvi) (b) 2, 8, 1

(xvii) (a) Both (A) and (R) are true and (R) is the correct explanation of (A)

(xviii) (d) Na

(xix) (c) X - iodine, Y - bromine

(xx) (a) Both the statements are true

Question 2

(i) (a) Electron affinity (b) Electronegativity (c) Ionisation potential

(ii) (a) largest (b) more

Question 3

(i) (a) K > Ca > Mg > Al

(b) Be < C < N < O

(c) F > P > Si > Be

(ii) (a) Down a group the atomic size increases (more shells), so the nucleus holds the outermost electron less strongly and less energy is needed to remove it.

(b) In the solid state the ions are held in fixed positions and are not free to move, so no current flows.

(iii) (a) Beryllium > Carbon > Fluorine

(b) Acetic acid (1) < Sulphuric acid (2) < Phosphoric acid (3)

(c) Potassium < Sodium < Lithium

(iv) (a) Hydrogen

(b) Sulphur

(c) Chlorine

Question 4

The atomic number of two atoms 'X' and 'Y' are 14 and 8 respectively. (Do not identify X and Y.)

(i) (a) Period 3 (X is 2, 8, 4)

(b) XY_2 (X has valency 4, Y has valency 2)

(ii) (a) Chlorine

(b) Neon

Elements P, Q and R are in the same period of the modern periodic table. P readily loses its one valence electron to form a stable ion. Q shares its electrons in bonding but does not form ions easily. R has high electronegativity.

(iii) (a) P

(b) R

(c) R > Q > P

(iv) (a) Fluorine

(b) Sodium

Question 5

Given below are a few elements: Li, K, Ca, F.

(i) (a) F

(b) Ca

(c) K

(ii) (a) K < Na < Si < S < Cl

(b) F < O < N < C < B < Be < Li

(c) F < Cl < Br < I

An element X has atomic number 17.

(iii) (a) Period 3, Group 17

(b) Non-metal

(c) HX (hydrogen chloride, HCl)

The electron affinity of an element X is greater than that of element Y.

(iv) (a) X has more oxidising power than Y

(b) X is more electronegative than Y

(c) X is placed to the right of Y

Question 6

X [2, 8, 7] and Y [2, 8, 2] are two elements.

(i) (a) Y

(b) 3

(c) Y

Element 'X' forms an oxide with the formula X_2O_3 which is a solid with high melting point.

(ii) 1. (c) Al

2. It has 3 electrons in the outermost shell (valency 3), like Al.

L, M and N are three elements with atomic numbers 13, 7 and 10 respectively. (Answer using only the alphabets given; do not identify the elements.)

(iii) (a) M

(b) N

(c) L

An element 'P' has 3 valence electrons in its L-shell and an element 'Q' has 7 valence electrons in its M-shell.

(iv) (a) Period 3, Group 17

(b) less

Question 7

In the reactivity series of metals, Z is a metal above hydrogen and its oxide has the formula ZO. It was found that ZO reacted with water and formed a hydroxide as the product.

(i) (a) 2

(b) Group 2 (alkaline earth metals)

(c) $Z(OH)_2$

CHAPTER 2: CHEMICAL BONDING

Question 1

(i) (c) Two lone pairs of electrons

(ii) (c) R

(iii) (b) Sodium chloride

(iv) (b) Ammonia

(v) (b) H_3O^+

(vi) (c) Both (A) and (R) are true, and (R) is the correct explanation of (A)

Question 2

(i) (a) Coordinate bond (b) Lone pair (c) Polar covalent bond

(ii) (a) bad (b) HCl

(iii) (a) 3 (b) 1 (c) 2 (d) 5 (e) 4

(iv) (a) 2 (b) 3 (c) 4 (d) 5 (e) 1

Question 3

(i) (a) Carbon shares one electron pair with each of four hydrogen atoms: 4 shared pairs on C (octet); each H has a duplet.

(b) Each N atom shares three electron pairs (triple bond, N≡N) and keeps one lone pair; each N has an octet.

(ii) (a) Their outermost shell is already complete (duplet or octet), so they neither lose nor gain electrons.

(b) Covalent compounds are made of molecules held by weak intermolecular forces, which need little energy to overcome.

(iii) (a) non-polar

(b) oxygen

(iv) (a) N shares three pairs with three H atoms and donates its lone pair to H^+ (coordinate bond): four shared pairs around N, positive charge on the ion.

(b) N≡N: three shared pairs, one lone pair on each N.

Question 4

(i) (a) $MgCl_2$: Mg (2, 8, 2) gives one electron to each of two Cl atoms, giving Mg^{2+} and two Cl^- each with 8 electrons.

(b) HCl: one shared pair between H and Cl; Cl keeps three lone pairs.

(c) NH_4^+ (ammonium ion): N shares three pairs with three H and donates its lone pair to H^+ (coordinate bond).

(ii) (a) Equal volumes of all gases, under the same conditions of temperature and pressure, contain the same number of molecules.

(b) A covalent bond in which the shared pair of electrons is contributed by only one of the two bonded atoms.

(iii) (a) sharing

(b) NH_3

(c) solid

(iv) (a) O shares two pairs with two H atoms and donates a lone pair to H^+ (coordinate bond); one lone pair remains on O; positive charge on the ion.

(b) C shares one pair with each of four H atoms.

Question 5

(i) (a) 1. False: the cation attains the configuration of the noble gas that comes before it (for example Na^+ is like neon).

2. As an ionic compound it conducts electricity only in the molten state or in aqueous solution, not in the solid state.

(b) 22 g ÷ 44 g/mol = 0.5 mol

(ii) O shares two pairs with two H atoms and donates one of its lone pairs to H^+ (coordinate bond); one lone pair remains on O; positive charge on the ion.

(iii) N shares three pairs with three H atoms and donates its lone pair to H^+ (coordinate bond), giving four shared pairs around N and a positive charge.

A compound commonly used in the laboratory has formula H_2P. H represents hydrogen, P represents a non-metal.

(iv) P is a Group 16 non-metal with 6 valence electrons: it shares one electron with each of two H atoms, giving two shared pairs and two lone pairs on P (octet); each H has a duplet.

Question 6

(i) Formula P_2Q (P loses 1 electron each, Q gains 2).

Mass of one mole = 2 × 23 + 16 = 62 g

(ii) (a) N shares three pairs with three H atoms; the fourth N–H bond is formed when N donates its lone pair to H^+ (this is the coordinate bond, shown with an arrow from N to H).

(b) N≡N: three shared pairs and one lone pair on each N.

CHAPTER 3: STUDY OF ACIDS, BASES AND SALTS

Question 1

(i) (b) 2

(ii) (d) Both 2 and 4

(iii) (c) 3

(iv) (c) Sodium sulphate + Lead nitrate

(v) (d) CH_3COOH

(vi) (d) Phenolphthalein

(vii) (d) Carbonic acid

(viii) (a) Ammonium hydroxide

(ix) (a) 1

(x) (c) NaCl

(xi) (d) 6 to 2

(xii) (d) Potassium sulphite

(xiii) (b) Q

(xiv) (c) 3 and 4

(xv) (b) Sodium sulphide

Question 2

(i) (a) Basic salt (b) Normal salt

(ii) (a) alkaline (b) $FeCl_3$ (c) hydrochloric acid (d) acidic (e) NH_4Cl

(iii) (a) 4 (b) 1 (c) 5 (d) 2 (e) 3

Question 3

(i) (a) Acetic acid has one replaceable hydrogen ion per molecule (monobasic); sulphuric acid has two (dibasic).

(b) In an electrolyte the current is carried by ions; in a metallic conductor it is carried by free electrons.

(ii) (a) HCl is monobasic: it has only one replaceable hydrogen ion, so it can only form a normal salt.

(b) Across a period the nuclear charge increases while the number of shells stays the same, so the atom attracts the shared pair more strongly.

(iii) (a) False (isomers have the same molecular formula but different structural formulae)

(b) True

Chemicals provided: Lead nitrate, Copper, Hydrochloric acid, Lead oxide, Iron, Sodium carbonate, Copper carbonate, Chlorine. (Chemicals may be used more than once, if needed.)

(iv) (a) $Pb(NO_3)_2 + Na_2CO_3 \rightarrow PbCO_3 + 2NaNO_3$

(b) $CuCO_3 + 2HCl \rightarrow CuCl_2 + H_2O + CO_2$ (or $Cu + Cl_2 \rightarrow CuCl_2$)

(c) $Fe + 2HCl \rightarrow FeCl_2 + H_2$

Question 4

(i) (a) 3

(b) 4

(c) 1

Mohan has three solutions P, Q and R having a pH of 13, 5 and 2 respectively.

(ii) (a) Q and R

(b) P

(c) Q

(iii) (a) The volume occupied by one mole of any gas at STP, which is 22.4 litres.

(b) A salt formed by the complete replacement of all the replaceable hydrogen ions of an acid by a metal or ammonium ion (for example NaCl).

P, Q, R and S are the different methods of preparation of salts. P – Simple displacement; Q – Neutralisation by titration; R – Precipitation; S – Direct combination.

(iv) (a) R

(b) S

(c) Q

Question 5

(i) (a) $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (Pt catalyst, about 800 °C)

(b) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

(c) $FeS + 2HCl \rightarrow FeCl_2 + H_2S$

List: CaO, CO_2, $NaOH$, $Fe(OH)_3$, CO

(ii) (a) CaO

(b) CO_2

(c) $NaOH$

(iii) (a) Sodium carbonate effervesces with dilute HCl, giving carbon dioxide that turns lime water milky; sodium sulphate shows no reaction.

(b) On warming with calcium hydroxide, ammonium sulphate gives ammonia (pungent smell, turns moist red litmus blue); sodium sulphate gives no gas.

Mahesh prepared a basic solution X that has a pH 7. The atomic number of an element is 15.

(iv) (a) 1. pH decreases

2. pH increases

(b) Group 15 (electronic configuration 2, 8, 5)

Question 6

Chemicals provided: Sodium hydroxide solution, copper carbonate, zinc, hydrochloric acid, copper, dilute sulphuric acid, chlorine, iron.

(i) (a) $CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2$

(b) $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$

(c) $2Fe + 3Cl_2 \rightarrow 2FeCl_3$

Chemicals provided: Sodium hydroxide solution, copper carbonate, zinc, hydrochloric acid, copper, dilute sulphuric acid.

(ii) (a) $CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2$

(b) $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$

Rohit has solutions X, Y and Z that have pH 2, 7 and 13 respectively.

(iii) (a) X

(b) Z

(c) Y